

ABDALLAH RABIE

Backend Developer

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Summary

Backend developer building production-grade systems with Python, Django, and FastAPI. Skilled in designing scalable APIs, optimizing database performance, and applying caching and asynchronous processing to build reliable, concurrency safe distributed services with clean architecture.

Projects

TwitterFlow API | *Django, DRF, PostgreSQL, Redis, Celery*

- Developed a modular monolith social media API with clear domain boundaries (accounts, tweets, interactions) enabling scalable architecture.
- Reduced tweet-detail query fan-out from 152 to 3 queries and improved latency by 447ms by eliminating N+1 queries via optimized joins and prefetching.
- Designed a cache-aside feed generation system achieving a 32× throughput increase and 96.9% reduction in response time.
- Ensured concurrency-safe and idempotent interactions using atomic operations and database constraints under high write load.
- Improved system responsiveness by offloading background tasks to Celery and mitigating abusive traffic via rate limiting.

Fast Broadcast | *FastAPI, WebSockets, Redis, PostgreSQL*

- Designed a distributed real-time messaging system using Redis pub/sub enabling cross-server WebSocket communication and room-based fanout.
- Built scalable presence tracking (global and per-room) using Redis sets and counters to maintain accurate online state across multiple sessions and server instances.
- Implemented multi-device connection management with per-socket tracking and isolated lifecycle handling for concurrent sessions.
- Engineered an efficient broadcast pipeline publishing events once to Redis and routing only to subscribed room sockets, reducing unnecessary fanout.
- Ensured data integrity and consistency using database constraints (partial indexes) and concurrency-safe handling of WebSocket lifecycle operations.

E-commerce API | *Django, DRF, PostgreSQL, Redis, Celery, Stripe*

- Built a modular E-commerce API (cart, orders, payments) using Django with Redis caching and Celery for asynchronous processing.
- Prevented inventory overselling using transactional stock checks and row-level locking, ensuring consistency under concurrent purchases.
- Optimized order queries from 24 to 4 (83%) using eager loading and aggregation techniques.
- Offloaded cart state to Redis hashes ($O(1)$ + TTL), eliminating database writes and improving scalability under high load.
- Improved checkout reliability by decoupling Stripe webhook handling and enforcing order state transitions via a state machine.

Technical Skills

Languages: Python, C/C++, SQL

Backend: Django, Django REST Framework (DRF), FastAPI

Databases: PostgreSQL, MySQL, Redis

Infrastructure: Git, Linux, Celery, Docker

Core Concepts: REST API Design, Asynchronous Processing, Caching, Concurrency, Authentication (JWT), OOP, Data Structures & Algorithms

Education

Bachelor's Degree in Computer Science and AI, Fayoum University

September 2023 – Present

GPA: 3.5

Fayoum STEM School

2020 – 2023

GPA: 3.7

Languages

Arabic: Native

English: Professional working proficiency